OmegaHelix

Predicting Drug-Protein binding Affinity

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Problem Overview

Summary - Predict how active compound(drug)-target(protein) interactions are
Dataset -
~330,000 entries

- Drug
  - Represented by SMILES
- Target
  - Represented by UniprotID + Amino acid sequence
- Affinity
  - Represented by pIC50 measurement

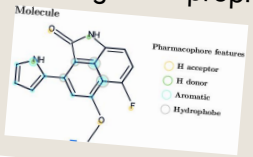
Drug Embedding

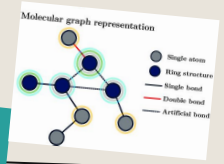
Molecular FP

Use RDKit's function to convert a SMILES into a Morgan Fingerprint (vector)

Molecule Graph

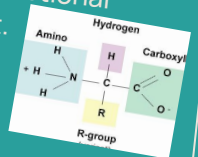
Use Chemprop to represent the molecule as a graph, and embed using Chemprop.





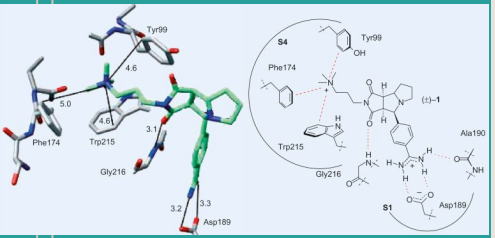
FP based on FG

Use RDKit's BRICS fragmentation to separate the molecule into functional groups. It's Morgan fingerprint is calculated based on the functional groups present.



FG Graph

Fragment the molecule into functional groups and use PyTorch Geometric to construct a graph from them.



Functional Group Fingerprinting

Explanation

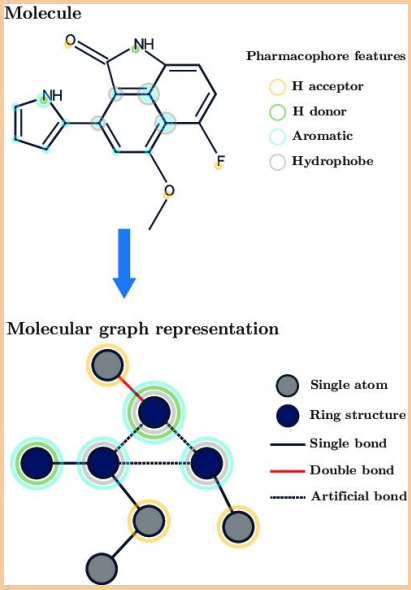
Atoms in themselves are not a great predictor of molecular interactions. Instead we can observe how they function as a group. In particular, functional groups often determine molecular properties, and we may be able to extract better insights from viewing the functional groups of a molecular rather than atoms individually

RDKit provides an algorithm known as BRICS, which separates a SMILES string into fragments based on bond strength, likely corresponding to a functional group. Each of these fragments are then fingerprinted. All of the molecule's fragments are averaged to create the 2048-dim vector.

Results

Model	Split	MSE	MAE	R2
XGBoost	Train	0.55	0.57	0.753
XGBoost	Valid	1.64	1.02	0.301
XGBoost	Test	1.81	1.05	0.244

- Fragment + fingerprint takes a very long time



Graph Representation of a Molecule

Explanation

A better way of understanding a molecule is a group of atoms and the interactions between them. A graph can provide a better depiction of a molecule, including bond strengths and angles. This allows the model to consider molecular substructure through distances from other nodes.

Chemprop offers a function that converts a SMILES to a graphical representation of a molecule.

Results

Model	Split	MSE	MAE	R2
XGBoost	Valid	1.54	0.99	0.298

Drug Embedding

Molecular FP

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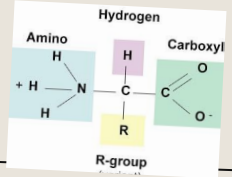
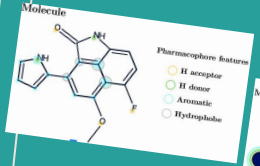
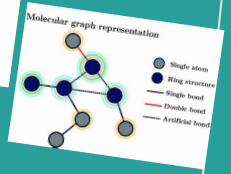


Molecule Graph

Use Chemprop to represent the molecule as a graph, and embed using its Chemprop.





FG Graph

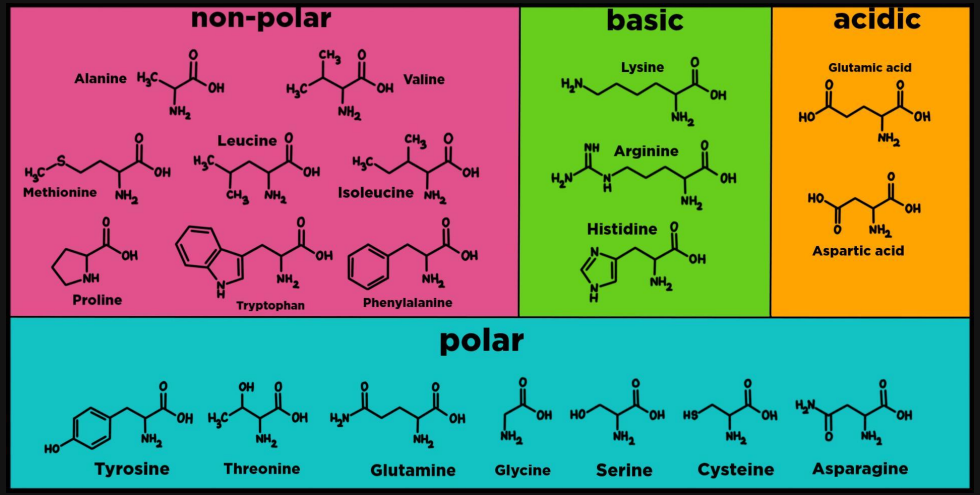
Fragment the molecule into functional groups and use PyTorch Geometric to construct a graph from them.

Protein Embedding

Input: Uniprot ID + Amino Acid Sequence

ProtTrans

1024-dimensional vector



Train
Eval
Test

Random split

Pros:

- Easier learning
- Lower variance

Cons:

- **Data leakage**
  - Inflates R^2

Protein split

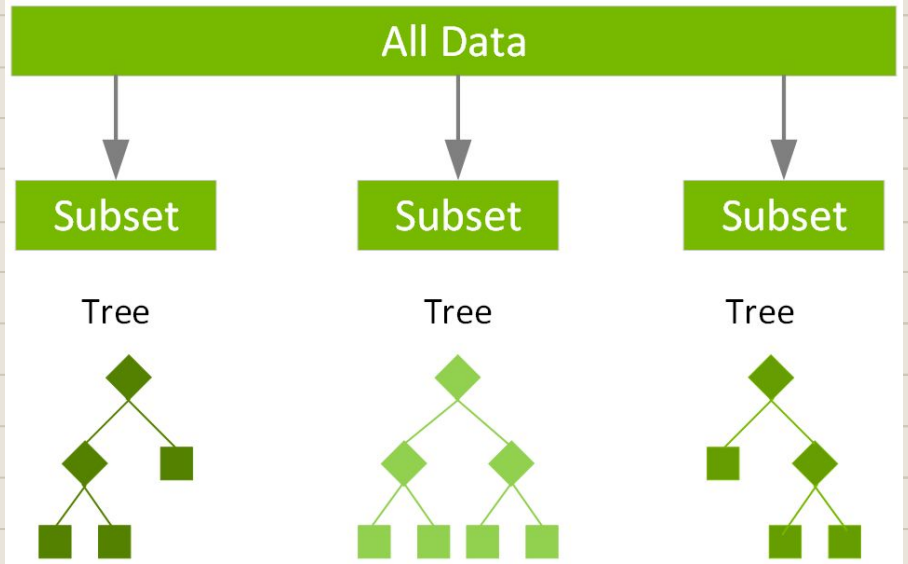
Pros:

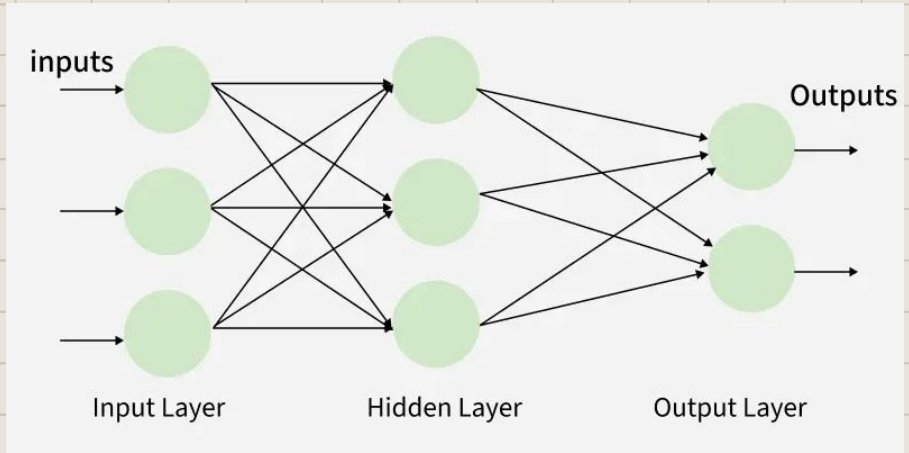
- Ensures proteins in training split don't appear in validation /test splits
- Tests generalization

Cons:

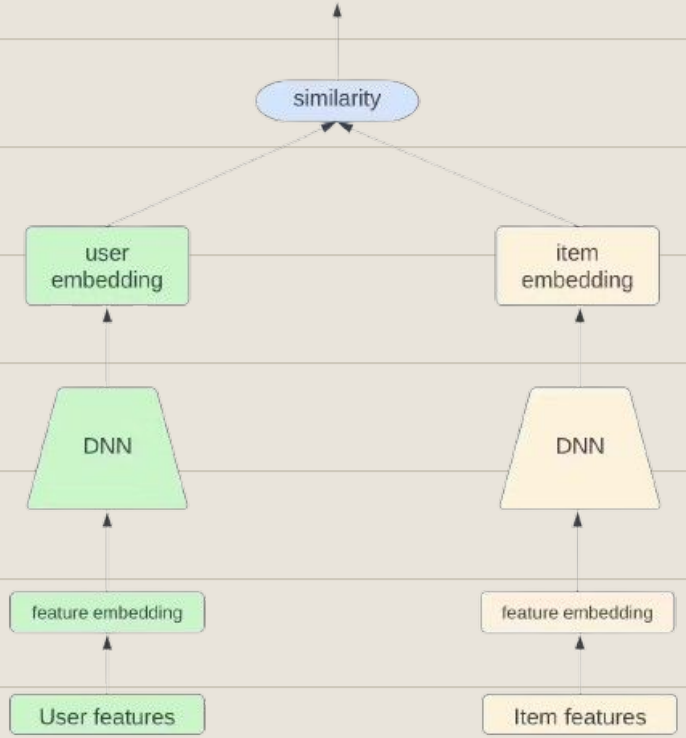
- Lower eval metrics
- Requires more data to perform

Model Selection



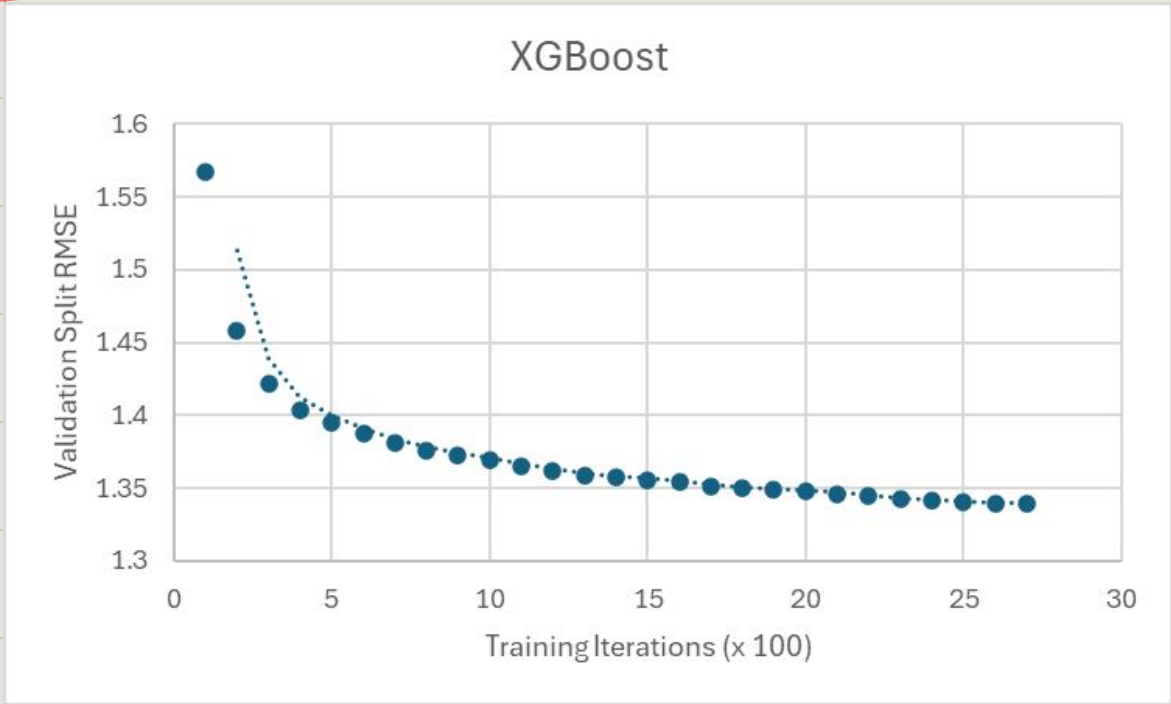


Multilayer Perceptrons



MLP valid: {'MSE': 1.6819, 'MAE': 1.0015, 'R2': 0.3176}
MLP test: {'MSE': 1.7546, 'MAE': 1.0195, 'R2': 0.1594}

XGBoost



Valid R^2: 0.2225
Test R^2: 0.1887

XGBoost vs Two Towers

	XGBoost	Two Towers
Valid R^2	0.2225	0.3176
Test R^2	0.1887	0.1594

Results

From running on test splits

THANK YOU!